Practical 1: Symmetry and the seven crystal systems

After completing this practical, you should be able to:

- Assign a unit cell to a 2D lattice
- Describe the orientation of planar features in lattices using Miller indices
- Recognise symmetry operators in 3D objects
- Deduce the crystal system of a 3D mineral from its macroscopic symmetry
- Use a polarizing microscope to examine a thin section under plane-polarized and cross-polarized light
- Identify cleavage and twinning in silicate minerals under the microscope

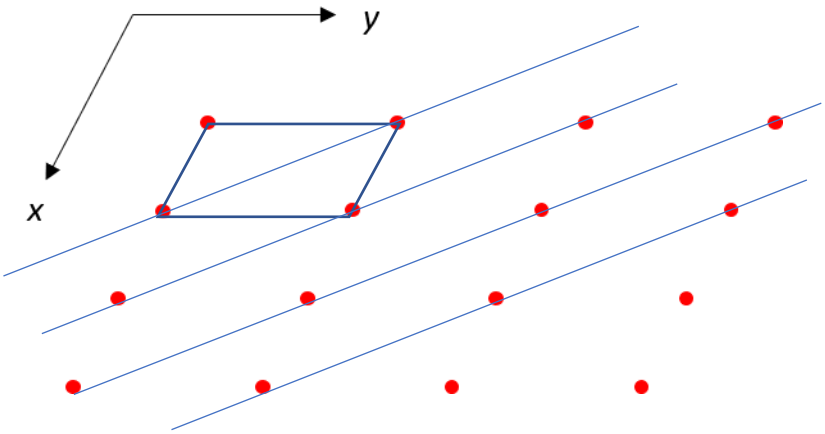
There are three parts to this practical. You can complete them in any order.

Part 1 (30 minutes): Unit cells and Miller indices

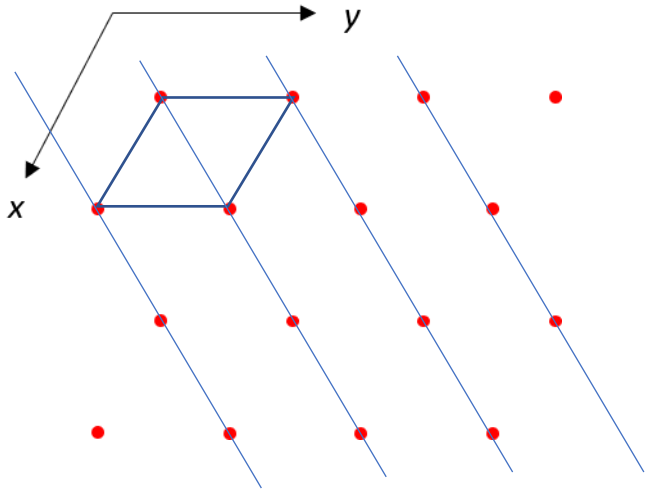
Below is a series of 2D lattices. For each lattice, draw a suitable unit cell, and measure the lattice parameters that are visible in the image (if you are working with a printed handout, you can measure the lattice dimensions; if you are working from a screen then you may need to estimate). *Hint: Remember to look at which crystallographic axes are shown on the diagram!*

For each lattice, draw the set of planes that correspond to the Miller indices specified.

i) Lattice parameters:
{110} planes

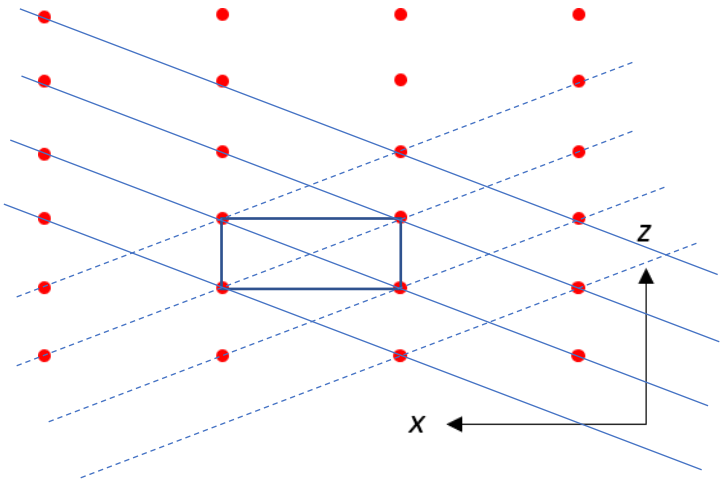


ii) Lattice parameters:
$\{1\bar{1}0\}$ planes



iii) Lattice parameters:
$\{101\}$ and $\{10\bar{1}\}$ planes

{101} in dashed lines, $\{10\bar{1}\}$ in solid lines



Part 2 (30 minutes): Symmetry in 3D objects

When minerals grow under ideal conditions, they develop external crystal faces. The characteristic external shape of a mineral is referred to as the *habit* (or crystal habit). The habit describes both the general shape and the crystallographic form, and also how well developed that form is. Importantly, the habit must conform to the underlying symmetry of the crystal. Recognising the habit can often help in identifying a mineral.

Work with the 3D crystal models. Select two or three models to take back to your desk. For each model, you should aim to:

i) Elements of rotation, reflection and inversion symmetry. Write down all the symmetry operators you identify (e.g. 3 diads, 3 mirror planes, centre of inversion).

ii) Examine how the symmetry operators are related to each other (e.g. the mirror planes are perpendicular to the diads).
iii) Using your symmetry classification, assign the model to one of the seven crystal systems.

Once you have finished, return the models to the tray at the front of the class. Briefly look through the rest of the models and check that you can identify at least one object that belongs to each of the seven crystal systems.

Part 3 (60 minutes): Cleavage and twinning in silicate minerals

There are three objectives to this exercise:

1) To become familiar with using a petrological microscope to examine a thin section;
2) To refresh your knowledge of common rock-forming silicate minerals;
3) To examine examples of cleavage and twinning in silicate minerals.

There are four thin sections, and you can look at them in any order. You should aim to make careful observations of least two of the thin sections. Make sure you have looked at each thin section by the end of the class, even if you do not have time to examine each one to the same level of detail.

Use the questions below to guide your observations. It is good practice to make labelled sketches as you go along.

RFM7

This rock is a hornblende syenite. The strongly coloured mineral is **hornblende**.

Describe and sketch the characteristic shape of the hornblende. You should find that there are at least two different characteristic shapes that represent slices at different angles through the 3D crystal.

You should be able to identify elongate sections with one cleavage trace, and regular near-octagonal sections with two cleavage traces. The crystal shapes are euhedral – that is, they are close to the ideal crystal habit. The 3D habit of hornblende is prismatic or acicular.

Explore the thin section and locate a twinned hornblende crystal. Is the twinning best seen in plane-polarized or cross-polarized light? Does hornblende show simple twinning (i.e., one visible twin plane) or lamellar twinning (i.e., multiple twins in the same crystal)?

Hornblende usually has simple twinning, but you may occasionally find a crystal with two twin planes.

Locate a hornblende crystal that shows two cleavage traces. Are the cleavage traces best seen in plane-polarized or cross-polarized light? What is the angle between the cleavage traces?

Cleavage is usually best seen in PPL, but should also be visible in XPL. Hornblende has two cleavages and the angle between them is approximately 60°

What is the relationship between the twin planes and the cleavage planes? Do they cross-cut each other? If so, at what angle?

In the two-cleavage section, the twin plane bisects the cleavage. In the one-cleavage section, the twin planes should be parallel to the cleavage.

What other rock-forming minerals can you identify in this thin section?

You will probably be able to recognise alkali feldspar and plagioclase. The thin section also contains apatite, which we will look at again in a later practical.

13415 *Belhelvie igneous complex, Aberdeenshire*

This rock is an orthopyroxene dolerite. The most abundant mineral in this sample is **plagioclase**.

Describe and sketch the characteristic shape of the plagioclases.

You should find that some of the plagioclases have regular rectangular shapes – they are euhedral. You will probably also notice some plagioclases with rather irregular shapes – these could be described as subhedral. In this rock, the 3D habit of plagioclase is tabular.

What planar features can you identify in the plagioclases? Are they visible in plane-polarized or cross-polarized light? Is this cleavage or twinning?

Under PPL, plagioclase is almost feature-less. Under XPL you should observe multiple lamellar twins that go extinct (black) at different points as you rotate the stage.

What other rock-forming minerals can you identify in this thin section? Do any of them show cleavage or twinning?

There are two minerals that are slightly coloured under PPL: clinopyroxene and orthopyroxene. Well done if you managed to tell them apart! (You will get more practice at this as the course continues.) Clinopyroxene and orthopyroxene both have two cleavage traces at 90° to each other, although depending on how the crystal has been sliced you may only be able to see one of the cleavages.

You should also notice that there is an opaque oxide mineral. This is probably magnetite.

SW

This rock is a biotite muscovite granite. The strongly coloured mineral is **biotite**.

Describe and sketch the characteristic shape (habit) of the biotites.

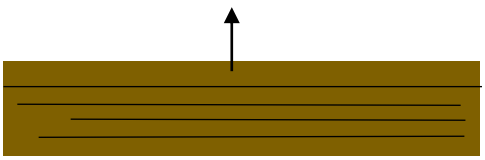
Biotite crystals are typically thin and elongate; the planar feature runs along the length of the crystal, and the short edges are ragged. You may also see some sections that are nearly hexagonal (although these are very rare in this sample). Biotite is usually described as having a platy habit.

What planar features can you identify in the biotite? Are they best seen in plane-polarized or cross-polarized light? Is this twinning or cleavage?

The planar feature is cleavage and is best seen in PPL.

The orientation of the planar features is {001}. Use this information to mark the direction of the crystallographic z axis on your sketch.

The {001} plane cuts the z axis, which means that the direction of the z axis must be perpendicular to the cleavage traces:



In plane-polarized light, rotate the thin section until the planar features are oriented east-west along the cross-hairs. What colour is the biotite in this orientation? What changes when you rotate the stage so that the planar features are oriented north-south?

When the cleavage is oriented E-W, the colour is dark brown. When the cleavage is oriented N-S, the colour is straw yellow. This colour variation is called pleochroism. We will examine this phenomenon more closely in Lesson 2.

What other rock-forming minerals can you identify in this thin section? Do any of them show cleavage or twinning?

You may be able to recognise plagioclase, alkali feldspar, quartz, and muscovite. These are all colourless in PPL.

R1 *Rum layered intrusion, Inner Hebrides, Scotland*

This rock is an olivine dolerite. The most abundant mineral is **olivine**.

Describe and sketch the characteristic shape (habit) of the olivines.

Most of the olivines in this thin section don't have obvious crystal faces – instead, they appear to be rather rounded, and should be described as anhedral. Note that this is not the ideal crystal habit of olivine!

Notice that the olivines have curving fractures cutting through the crystals. Are they visible in plane-polarized or cross-polarized light? Is this cleavage, or something else? How do you know?

Curving fractures are common in olivine. They are prominent in PPL, and visible in XPL. They cannot be cleavage, because cleavage is defined to be a planar feature – so in thin section they should always appear as linear features (or very close to linear). Olivine does actually have three cleavage traces at 90° to each other, but they are usually very faint or invisible in thin section.

What other rock-forming minerals can you identify in this thin section?

You may be able to identify clinopyroxene and plagioclase surrounding the olivines. You should also notice an opaque oxide mineral – this is spinel.

General advice for describing minerals in thin section

When you describe a mineral's appearance in thin section, it is a good idea to look for the following features (note that this is not an exhaustive list):

- Are there any cleavage traces? If there is more than one set of cleavage traces, what angle do they make with one another?
- Does the mineral show twinning? If yes, is it simple twinning (i.e. there is only one visible twin plane) or lamellar twinning (i.e. there are multiple sets of twin planes in the same crystal)?
- If you don't identify any planar features in the mineral, it is a good idea to record this observation

These kinds of observations would be required if you needed to identify an unknown mineral in a thin section. (And in an exam-style question you will pick up marks for recording these observations, even if you are not able to identify the mineral.)

Bonus question

If you've finished and would like a challenge, try this question. You can also use it for revision.

A tetragonal mineral with lattice dimensions a = 12 Å, c = 6 Å has a body-centred lattice; a mirror plane with Miller index (001); and two perfect cleavages in the orientations {100} and {010}. Draw projections of the (100) and (001) sections through this mineral (i.e., looking down the x and z axes), showing the shape of the unit cell, the mirror and cleavage planes, and the positions of the lattice points.

Solutions available on request! Please email me if you attempted this question and would like feedback on your work.